Day 1 — Course intro & what is R?

Monday, May 18, 2026

> **ℹ Today**
>
> **Before class**
>
> - Read the syllabus (linked from Canvas).
> - Read *Data Computing* §1.
>
> **Plan for today**
>
> - Course logistics
> - Introduction to R and RStudio
> - Tidy data
> - R Demo
>
> **Helpful links**
>
> - Syllabus · AI policy · Schedule · *Data Computing* §1

1. Course Info & Introductions

Summer Session 1: May 18, 2026 – June 26, 2026

- **Class time:** M/T/W/Th/F, 9:35 a.m. to 10:50 a.m.
- **Class location:** Huck Life Sciences Building 007
- **Instructor:** Charles Costanzo
  - Email: cgc5478@psu.edu
  - Office: Thomas 301 (Office Hours TBD)
- **Teaching Assistant (TA):** Xinyue Wang

  - Email: xpw5228@psu.edu

> Icebreaker: name, major, your favorite song right now.

Learning Goals

This course will teach you how to:

- **Use R**: Program in the R *language* effectively and use the R *software environment* to efficiently compute with data.
- **Wrangle Real Data**: Access, clean, join, and format data from various sources.
- **Visualize Information**: Create graphical and descriptive summaries of data.
- **Build Reproducible Workflows**: Write well-documented code using tools like RStudio, RMarkdown, and Git/GitHub.

> No previous programming background is required for this course.

> ! This is **not** a course on general purpose programming.

Course logistics

- **Six weeks is short.** Daily pace is faster than a 15-week course. Falling a week—or even a day—behind is more serious than spring/fall courses.
- **Attendance matters.** Lots of in-class graded work (activities, quizzes, oral checkpoints).
- **Timely communication is critical.** Reach out to me **as soon as possible** if something comes up that necessitates you missing class, having a deadline extended, etc.
- **Assignment Weights:**

Readings	Quizzes	In-class activities	Homework	Exam	Course project
10%	15%	25%	10%	20%	20%

- **Grading Scheme:**

A	**A-**	**B+**	**B**	**B-**	**C+**	**C**	**D**	**F**
93%+	90%	87%	83%	80%	77%	70%	60%	$<$ 60%

AI policy

- AI is allowed on homework, take-home exam, project **with disclosure**. *Not* allowed on in-class work.
- Disclosure is graded on homework, but it is **not** an integrity issue if you skip it (you will just lose points.)
- Lying about AI use *is* an integrity issue.
- Oral project checkpoints are how I verify you understand your own work.
- Practical implication: AI is a study aid, not a substitute. The course is designed so that students who lean on it too heavily will struggle on the in-class portions.

Discussion Questions

- How have you used AI in past courses?
- What are your initial thoughts on the policy?

What is R?

R is a **programming language** *and* a **software environment** for statistical computing and graphics.

- Designed by statisticians for the purpose of data analysis.
- First version created in August 1993 (almost 33 years ago).
- Has grown rapidly in popularity and is widely used in academia and industry by statisticians, data analysts, and data scientists alike.
- Enormous number of user-contributed **packages**, which are add-ons that extend the functionality of R.
  - Currently the Comprehensive R Archive Network (CRAN) package repository features 23,670 available packages.[1]

Figure 1: Hex stickers from the rstudio/hex-stickers repository.

Some Examples of Said R Packages

R vs. RStudio

- **R** is a computer language that performs computations based on human-written instructions.
- **RStudio** is an integrated development environment (**IDE**) for R programming. It combines the R console with supplementary tools for more efficient R programming.

(a) **R:** Do not open this

(b) **RStudio:** Open this

Figure 2: R and RStudio logos.

[1] As of May 15, 2026.

Tidy Data ≠ Neat Data

> This table is easy for humans to read. What might make it tricky for a computer to read?

	Week Of 05-04-2026 to 05-11-2026				Yearly Total 2026			
	To Democratic		To Republican		To Democratic		To Republican	
COUNTY	R	OTH	D	OTH	R	OTH	D	OTH
ADAMS	5	4	4	3	71	93	74	102
ALLEGHENY	47	125	23	31	786	1,323	810	389
ARMSTRONG	2	0	1	1	23	28	65	26
BEAVER	2	15	3	1	77	127	150	55
BEDFORD	0	1	0	1	13	14	21	16
BERKS	15	37	14	16	337	494	422	335
BLAIR	5	7	0	3	60	83	62	52
BRADFORD	1	2	1	2	32	27	40	45
BUCKS	32	94	15	20	626	1,054	550	557
BUTLER	3	5	0	3	131	170	101	111
CAMBRIA	1	2	3	0	82	71	174	62
CAMERON	0	0	0	0	1	1	7	0
CARBON	5	11	4	5	34	70	54	43
CENTRE	2	14	3	4	88	130	48	75
PHILADELPHIA	60	313	66	29	1,076	3,163	1,173	489
VENANGO	2	2	0	0	28	35	30	27
WARREN	0	0	1	1	24	32	15	24
WASHINGTON	5	7	7	3	112	142	188	111
WAYNE	1	1	3	2	22	35	41	39
WESTMORELAND	8	20	12	8	211	238	302	168
WYOMING	0	2	3	0	22	22	27	17
YORK	24	60	10	22	415	656	325	393
Totals:	**512**	**1,348**	**334**	**322**	**9,823**	**15,604**	**9,631**	**7,295**

Figure 3: A spreadsheet showing party change applications in PA. Source: PA Dept. of State (May 4–11, 2026).

Two simple rules for Tidy Data:

1. **Consistent Cases (Rows)**: Every row represents the same underlying attribute
2. **Uniform Variables (Columns)**: Every column contains the same type of value.

PA Voter Table - Tidied Up

```r
# 1. Load data and provide clean column names to handle the nested headers
raw_data <- readxl::read_excel("data/currentvotestats.xlsx",
                     sheet = "Party-to-Party(2026)",
                     skip = 3, # Skip the human-readable headers
                     col_names = c("county",
                                   "week_dem_R", "week_dem_OTH",
                                   "week_rep_D", "week_rep_OTH",
                                   "year_dem_R", "year_dem_OTH",
                                   "year_rep_D", "year_rep_OTH"))

# 2. Clean and Pivot
tidy_voter_data <- raw_data %>%
  filter(!is.na(county), county != "Totals:") %>% # Rule 1: Consistent Cases
  pivot_longer(
```

```r
    cols = -county,
    names_to = c("period", "target_party", "origin_party"),
    names_sep = "_",
    values_to = "count"
  )

knitr::kable(head(tidy_voter_data %>% filter(county == "CENTRE")))
```

county	period	target_party	origin_party	count
CENTRE	week	dem	R	2
CENTRE	week	dem	OTH	14
CENTRE	week	rep	D	3
CENTRE	week	rep	OTH	4
CENTRE	year	dem	R	88
CENTRE	year	dem	OTH	130

Demo: what R can do

Live demo — students just watch, no need to follow along.

Goals: - Show R is a real programming environment, not a calculator. - Walk through a complete small analysis: load, clean, visualize, summarize. - Make it look easy.

Narrate at a high level — don't get into syntax, they'll learn that.

Candidate datasets: - `palmerpenguins::penguins` — clean and visually striking (current pick) - Tidy Tuesday "horror movies" or similar topical - NYC airbnb pricing

```r
library(tidyverse)
library(palmerpenguins)
```

```

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

    penguins, penguins_raw
```

```r
# What does the data look like?
glimpse(penguins)
```

```
Rows: 344
Columns: 8
$ species           <fct> Adelie, Adelie, Adelie, Adelie, Adelie, Adelie, Adel~
$ island            <fct> Torgersen, Torgersen, Torgersen, Torgersen, Torgerse~
$ bill_length_mm    <dbl> 39.1, 39.5, 40.3, NA, 36.7, 39.3, 38.9, 39.2, 34.1, ~
$ bill_depth_mm     <dbl> 18.7, 17.4, 18.0, NA, 19.3, 20.6, 17.8, 19.6, 18.1, ~
$ flipper_length_mm <int> 181, 186, 195, NA, 193, 190, 181, 195, 193, 190, 186~
$ body_mass_g       <int> 3750, 3800, 3250, NA, 3450, 3650, 3625, 4675, 3475, ~
$ sex               <fct> male, female, female, NA, female, male, female, male~
$ year              <int> 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007, 2007~
```

```r
# Quick summary by species
penguins |>
  group_by(species) |>
  summarize(
    n             = n(),
    mean_mass_g   = mean(body_mass_g,    na.rm = TRUE),
    mean_bill_mm = mean(bill_length_mm, na.rm = TRUE)
  )
```

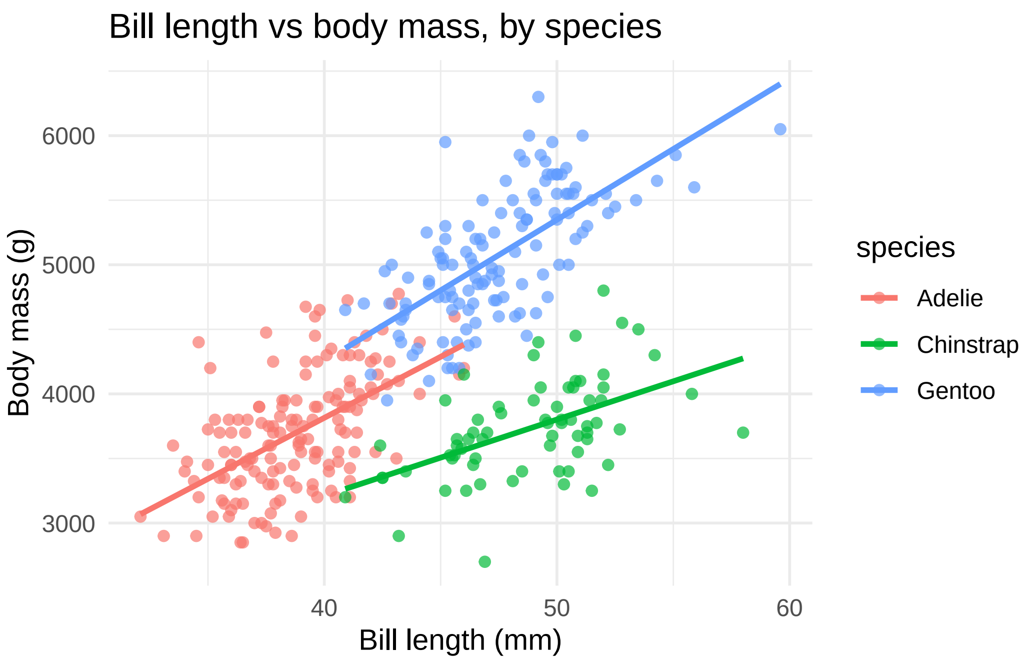
```
# A tibble: 3 x 4
  species       n mean_mass_g mean_bill_mm
  <fct>     <int>       <dbl>        <dbl>
1 Adelie      152       3701.         38.8
2 Chinstrap    68       3733.         48.8
3 Gentoo      124       5076.         47.5
```

```r
# A nice plot
penguins |>
  ggplot(aes(x = bill_length_mm, y = body_mass_g, color = species)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "Bill length vs body mass, by species",
    x = "Bill length (mm)",
    y = "Body mass (g)"
  ) +
  theme_minimal()
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Setup checklist — due by Tuesday

Install R and RStudio

- R: https://cloud.r-project.org
- RStudio: https://posit.co/download/rstudio-desktop

> ⚠ Mac users
>
> Install **XQuartz** too — some packages we'll use later in the course need it.

Configure RStudio

- `Tools > Global Options... > General`
- Restore .RData into workspace at startup: **Unchecked**
- Save workspace to .RData on exit: **Never**

Assignments due Tuesday

- HW 0: Setup confirmation
- Reading Quiz 1 on *Data Computing* §2.1–2.3
- Read *Data Computing* §2.1–2.3 before class

Wrap-up & looking ahead

Tomorrow: Real R syntax (variables, functions, vectors)

Due Tuesday before class:

- HW 0: Setup confirmation
- Reading Quiz 1 on Canvas
- Read *Data Computing* §2.1–2.3

Reference links

- Syllabus · AI policy · Schedule · *Data Computing*

> Reach out
>
> Stuck on setup? Email me (cgc5478@psu.edu) or Xinyue (xpw5228@psu.edu) before Tuesday — don't wait until class.